Concept of prevention

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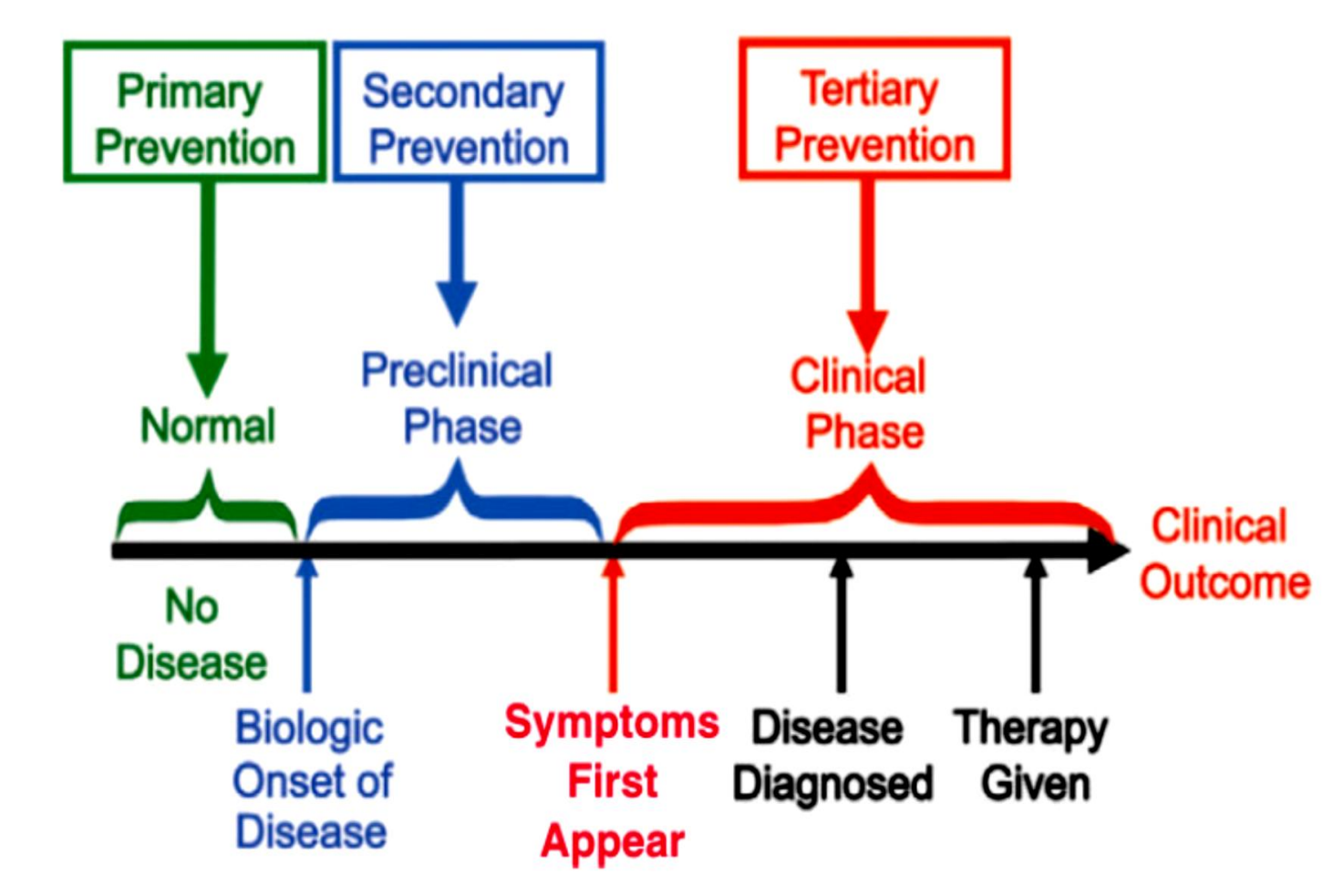
Intended learning out come

By the end of the lecture the student will be able to:

- **Identify the different levels of prevention.**
- **Describe the principles of prevention of communicable diseases.**

- Objective:
  - Oppose the disease process
  - Preserve health
  - Promote health
  - Prolong life
  - Minimize suffering by preventing disease occurrence
- three levels of prevention
  - primary,
  - secondary and
  - tertiary.



1-Primary prevention:

- This is the measure undertaken in the period of prepathogenesis (i.e. before the development of the disease) which removes the possibility of occurrence of the disease.
- Especially, it is all the more important in those diseases, for which no treatment is available, e.g. AIDS, cancer, rabies, carries tooth, etc.
- Primary prevention can be adopted by two modes of intervention—namely Health promotion and Specific protection.

Health promotion

- This consists of ‘general measures’, which will strengthen the individual/host and prevents the occurrence of the disease by interrupting the interaction among the three factors of epidemiological triad.

The various measures of health promotion are:

- Health education (on personal hygiene, oral hygiene, nutrition education, life-style, etc.). Sex education.
- Adequate nutrition.
- Improvement in the environmental sanitation (such as control of insects, provision of protected water supply, sanitary disposal of sewage, etc.).
- Promotion of breastfeeding and proper weaning.

- Family planning and spacing of births.
- Genetic counseling (premarital and marriage counseling).
- Efficient antenatal care and postnatal care.
- Recreation facilities (sports, games, cultural activities, etc.).
- Improvement in the literacy
- Yoga exercises and meditation.

Specific protection

- This consists of ‘specific measures’, which prevent specific diseases. The various measures are:
  - Immunization against vaccine preventable diseases
  - Silver nitrate or penicillin eye drops against ophthalmia neonatorum
  - Condom against AIDS
  - Use of specific nutrients (Vitamin A against nutritional blindness, Iron and folic acid against nutritional anemia, iodized salt against iodine deficiency disorders)

- Masks against pneumoconiosis
- Ear plugs against noise induced deafness
- Lead apron against radiation hazards
- Barrier cream against occupational skin cancer
- Avoidance of allergens and carcinogens

- Sterilization procedures of surgical instruments
- Pasteurization of milk
- Traffic signals against road accidents
- Quality control of foods, drugs and cosmetics, etc.

- These measures need to be applied in specific situation for specific groups. They are more concrete and effective.

PRIMORDIAL PREVENTION

- A **primary level of prevention** focused on **noncommunicable** diseases (e.g., obesity, hypertension, diabetes, cancer, coronary artery disease).
- Involves elimination or modification of **risk factors** before they become established.
- Etiology discussed in terms of risk factors rather than a single causative agent.

- **Approaches**

1. **Population Strategy**
   - Directed at the whole population, regardless of risk level.
   - Promotes lifestyle changes through health education from childhood.
   - Effects are long-term, visible after years or decades.
2. **High-Risk Strategy (Individual Strategy)**
   - Directed at individuals at high risk of disease.
   - High-risk groups identified through screening procedures.
   - Provides targeted interventions to prevent disease onset.

2- Secondary prevention:

- This is the measure undertaken in the **early stage, after the onset of the disease** or even much before the development of permanent pathology in the individual.
- Aims to reduce the impact of disease or injury that has already occurred.
- Focuses on **early detection** and treatment during the preclinical phase (after disease develops but before symptoms appear).
- Encourages strategies to prevent recurrence and programs to restore health and function to prevent long term problems .
- Aims to reduce the impact of disease or injury that has already occurred.

- Examples
  - Regular exams & screening tests (e.g., mammograms for breast cancer).
  - Daily low-dose aspirin, diet & exercise programs (prevent further heart attacks/strokes).
  - Modified work conditions for safe return of injured/ill workers

Early Diagnosis and Prompt Treatment

This helps in the following ways:

- Helps in recovery from the disease (restoration)
- Reduces the duration of illness in the individual
- Minimizes the suffering
- Prevents the development of complications
- Prevents further spread of the diseases in the community
- Prevents or postpones the death of the individual.

3- Tertiary prevention:

- This is the measure undertaken, when the disease process is **sufficiently advanced**, (i.e. in the late pathogenesis phase).
- Tertiary prevention can be adopted by two modes of intervention
  - disability limitation and
  - rehabilitation.

Disability limitation:

- This means **limiting** the development of further disability in the individual by giving intensive or aggressive treatment, when the patient comes in the advanced stage of the disease.
- The objective is to prevent the transition from impairment to handicap.
- The sequence of events in a disease process are:
  - Disease → Impairment → Disability → Handicap

- **Impairment** : This means **defect** in the structures and function of an organ or a part of the body.
- **Disability** : This **means inability to carry out certain routine, expected activities**, considered normal for the age, sex etc., due to impairment.
- **Handicap** : This means experiencing **disadvantage** in the life and not able to play the role, expected out of her/him, resulting from the impairment or disability.

Rehabilitation

- Rehabilitation has been defined as the ‘combined and coordinated use of medical, social, educational and vocational measures for training and retraining the individual to the highest possible level of functional ability”

The different aspects of rehabilitation are::

Physical (medical) rehabilitation	Restoration of function
Vocational rehabilitation	Restoration of earning capacity
Social rehabilitation	Restoration of relationship in the society
Psychological rehabilitation	Restoration of personal dignity and confidence

Examples of rehabilitation are:

- Establishing the schools for the blind
- Providing aids for crippled, such as artificial limb, crutches, wheel chair, hearing aid, etc.
- Reconstructive surgery in leprosy
- Graded exercises in paralysis
- Intraocular implantation of lens among cataract patients.

Thank you